

Job Opening

Research Engineer in Machine Learning

for polymer-property prediction

Contract	Six-month, full-time fixed-term position
Starting date	From January 2027, exact date to be agreed
Location	LITIS, INSA Rouen Normandie, Madrillet Campus
Objective	Consolidate the benchmark and submit a scientific article

Research Topic

The ANR **OCTOPUSSY** project brings together machine learning, theoretical chemistry, and polymer science. The position focuses on predicting polymer glass-transition temperature (T_g) from enriched molecular representations: 2D graphs, conformer ensembles, quantum descriptors, and chemical language models.

The position builds on an original dataset and an established set of experiments. The central question is:

When do conformer ensembles and quantum descriptors genuinely improve T_g prediction compared with 2D representations and chemical language models?

The work will be carried out at LITIS in close collaboration with the team at the CARMEN chemistry laboratory.

Responsibilities

The successful candidate will contribute to:

- taking over and reproducing the existing experiments on the CRIANN computing centre;
- consolidating a shared benchmark of tabular models, GNNs, and chemical language models;
- analysing the existing results to assess the contribution of quantum descriptors, several conformer-aggregation strategies, and monomer, dimer, and trimer representations;
- documenting the dataset, computations, and code with a view to releasing them;
- actively contributing to the writing and submission of a scientific article.

The project builds on advanced work and completed experiments. It will favour controlled and interpretable comparisons rather than the development of a new architecture.

Candidate Profile

- Engineering degree, Master's degree, or PhD in computer science, machine learning, data science, cheminformatics, or a related field.
- Excellent command of Python and at least one deep-learning framework, preferably PyTorch.
- Experience in conducting quantitative experiments and evaluating models.
- Ability to take over and document an existing scientific codebase.
- Good command of written scientific English and interest in interdisciplinary collaboration with chemists.

Experience with RDKit, PyTorch Geometric, GNNs, Transformers, Slurm, or high-performance computing would be appreciated.

Environment and Conditions

LITIS provides a machine-learning research environment within INSA Rouen Normandie. The project provides access to a polymer dataset enriched with quantum calculations, existing learning pipelines, and computing resources at

CRIANN, Jean Zay, and the local LITIS cluster. Regular exchanges with chemists will connect numerical results with their molecular interpretation.

Salary	According to the candidate's profile and experience and the salary scale applicable at INSA Rouen Normandie.
Location	Saint-Étienne-du-Rouvray, Rouen metropolitan area, France.
Application timeline	Preferred start in January 2027; applications reviewed on a rolling basis until the position is filled.

Why Join the Project?

- A timely scientific question at the intersection of machine learning, quantum chemistry, and polymer materials.
- An original dataset containing conformer ensembles and quantum descriptors.
- Models, data splits, and results already available for a rapid start.
- Direct collaboration with specialists in machine learning and chemistry.
- A concrete scientific-publication objective during the contract.

Application

Applications should include a detailed CV, a short cover letter outlining the candidate's suitability and availability, links to relevant work (publication, thesis or dissertation, code, or software repository), and contact details for one or two referees.

Send applications to: benoit.gauzere@insa-rouen.fr
Email subject: OCTOPUSSY 2027 Application – First name Last name

INSA Rouen Normandie is committed to professional equality, diversity, and the inclusion of people with disabilities.